Response to the Deputy Editor's Pre-Review Comments

Manuscript: ***Spatio-temporal wall-actuation modes and cycle-mean viscous dissipation in a lid-driven cavity***

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To: The Deputy Editor, *Physics of Fluids*.
Re: Revision requested before external review. The revised manuscript and this response are submitted through the PXP site by 21 November 2026.

Opening statement

We thank the Deputy Editor for confirming that the chosen article type is appropriate and for the pre-review comments. The manuscript has been revised to meet them. This letter answers each comment in the order it was given. The changes themselves are visible in the marked-up manuscript and are itemized, comment by comment, in the point-by-point response transmitted with the revision; they are not repeated here.

Comment 1

"Physics of Fluids publishes original theoretical, computational, and experimental work that advances the physical understanding of fluids, meaning the mechanisms and principles that explain how and why fluids behave as they do. As written, the manuscript is framed as a reproducibility analysis of a physics informed neural network controller for the lid driven cavity. Before it can be considered, the paper needs to be reorganized so that the fluid physics is its central contribution."

Response. The manuscript has been reorganized around the fluid physics. It now studies how the spatial and temporal structure of a wall actuation sets the viscous dissipation of the cavity flow it drives, and the physics-informed network appears only as the tool that generates the candidate actuations, which an independent multiple-relaxation-time lattice-Boltzmann solver then evaluates. The identification campaigns and the reproducibility audit no longer frame the paper.

Comment 2

"The physics to bring to the front is how the spatio temporal structure of the wall actuation sets the cycle-mean viscous dissipation of the cavity flow. The neural network branch selection, the regularization dependent state recovery, and the reproducibility audit can remain as supporting evidence, but they should not frame the work."

Response. The cycle-mean dissipation is now the quantity the paper compares, at a common quadratic actuation norm, for the stationary modes and the three identified branches. The distinction between the cycle-mean dissipation and the dissipation of the time-mean field, and the partition of the cycle-mean cost between mean field and fluctuations, are reported for each branch. The branch selection, the regularization-dependent state recovery and the reproducibility audit remain as supporting evidence in the supplementary material.

Comment 3

"The introduction and reference list should be repositioned toward the fluid physics literature on active control of the lid driven cavity, viscous dissipation and energy budgets in actuated cavity flows, and modal structure and state selection in driven-cavity flow, so that the contribution is set against fluid flow."

Response. The introduction and the reference list are now positioned in that literature: the cavity benchmarks, the active-control and cost-function work, the dissipation and energy-budget studies, and the modal structure and state-selection results. The antecedent identification study by one of us is cited where its results are used and is not restated.

Comment 4

"Please also reduce the number of figures to maximum of 15."

Response. The revised manuscript carries eight figures and four tables. The figures of the identification campaigns are no longer in the main text; those that support the comparison are in the supplementary material.

Comment 5

"Please revise the title to lead with the actuation mode and dissipation physics."

Response. The title is now *Spatio-temporal wall-actuation modes and cycle-mean viscous dissipation in a lid-driven cavity*, which leads with the actuation structure and the dissipation cost.

Comment 6

"Please revise your manuscript and submit a detailed, point-by-point response to the comments. Each response should include the original comment followed by a description of the corresponding revision made (or a justification if no change was made). The revised manuscript and response are due by 21-Nov-2026 via the PXP web site. The Editors will then make a decision as to the next step in the review process."

Response. A detailed point-by-point response is submitted with the revision. It reproduces every comment of the Deputy Editor and of the reviewers, each followed by the corresponding revision or by the justification where no change was made. The revised manuscript and the response are submitted through the PXP site by 21 November 2026.

Comment 7

"When submitting your revision, please upload two versions of your manuscript: a clean version labeled 'Article File' and a marked-up version labeled 'Marked-Up Manuscript,' which should clearly show all changes made from the previous submission."

Response. Both versions are uploaded: the clean manuscript as the Article File and the marked-up manuscript as the Marked-Up Manuscript, the latter showing every change against the previous submission. The same pair is provided for the supplementary material.

Closing

The revised manuscript, the marked-up manuscript, the supplementary material in both versions and the point-by-point response accompany this letter. We remain available to the Editors for the next step of the review process.

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Signed on behalf of all the authors

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